

Exercise of M670_5 (Continuous Distributions)

Question 1: Your professor's exam scores are normally distributed with a mean of 80 and a standard deviation of 8. What's the probability that you'll score above 90? (Are you aiming too high?) And what's the probability that you'll score below 60? (How many nights of studying would you need to avoid this?)

Answer:

Given the problem:

- Mean (μ) = 80
- Standard deviation (σ) = 8

We can calculate the Z-scores using the formula:

$$Z = \frac{X - \mu}{\sigma}$$

1. Probability of scoring above 90:

$$Z_{90} = \frac{90 - 80}{8} = 1.25$$

The probability of scoring above 90 is:

$$P(X > 90) = 1 - P(Z \leq 1.25) = 1 - 0.8944 \approx 0.1057$$

So, the probability of scoring above 90 is approximately 10.57%.

2. Probability of scoring below 60:

$$Z_{60} = \frac{60 - 80}{8} = -2.5$$

The probability of scoring below 60 is:

$$P(X < 60) = P(Z \leq -2.5) = 0.0062$$

So, the probability of scoring below 60 is approximately 0.62%.

Question 2: The flight time from Chicago O'Hare (ORD) to Purdue University (LAF) is approximately normally distributed with a mean of 60 minutes and a standard deviation of 8 minutes. What's the probability that a randomly selected flight has a duration:

1. Between 55 and 65 minutes?
2. Longer than 70 minutes?
3. Why are these results useful?
4. The middle 90% of flights will have durations between what two values?

Answer:

Let the flight time be the random variable $X \sim N(\mu = 60, \sigma = 8)$.

1. Probability between 55 and 65 minutes:

Standardize using the z -score:

$$z = \frac{X - \mu}{\sigma}$$

For $X = 55$:

$$z_1 = \frac{55 - 60}{8} = -0.625$$

For $X = 65$:

$$z_2 = \frac{65 - 60}{8} = 0.625$$

Using the standard normal table:

$$P(Z < 0.625) \approx 0.734$$

$$P(Z < -0.625) \approx 0.266$$

Therefore,

$$P(55 \leq X \leq 65) = P(Z \leq 0.625) - P(Z \leq -0.625) \approx 0.734 - 0.266 = 0.468$$

So, the probability is approximately **46.8%**.

2. Probability longer than 70 minutes:

Standardize:

$$z = \frac{70 - 60}{8} = 1.25$$

Using the standard normal table:

$$P(Z > 1.25) = 1 - P(Z \leq 1.25) = 1 - 0.8944 = 0.1056$$

So, the probability is approximately **10.6%**.

3. Why these results are useful:

- **For airlines:** They help in scheduling gates, crew, connecting flights, and managing operational risks.
- **For insurance and pricing:** Knowing the probability of unusually long flights (e.g., longer than 70 minutes) helps insurance companies price delay coverage and assess risk. It also helps airlines or travel agencies design delay compensation policies.
- **For passengers:** Knowing the typical flight duration range helps passengers plan arrivals, connections, and travel insurance with confidence.

4. The middle 90% of flights will have durations:

For the middle 90%, the lower and upper bounds correspond to the 5th and 95th percentiles of the normal distribution.

$$z_{0.05} \approx -1.645, \quad z_{0.95} \approx 1.645$$

Convert z-scores to actual flight times:

$$X_{\text{lower}} = \mu + z_{0.05} \cdot \sigma = 60 + (-1.645)(8) \approx 46.8$$

$$X_{\text{upper}} = \mu + z_{0.95} \cdot \sigma = 60 + (1.645)(8) \approx 73.2$$

Middle 90% interval: 46.8 minutes to 73.2 minutes

Question 3: Your total sleep time per night is normally distributed with a mean of 8 hours and a standard deviation of 1 hour. After several late nights during finals, you decide that you need to sleep at least as much as the top 10% of your normal nights. How many hours of sleep should you aim for? (Time to put your phone away!)

Answer:

- Mean (μ) = 8 hours
- Standard deviation (σ) = 1 hour

We need to find the value corresponding to the top 10%, which is the 90th percentile:

$$Z_{90\%} = 1.28$$

The corresponding sleep time is:

$$X = \mu + Z \cdot \sigma = 8 + 1.28 \cdot 1 = 9.28 \text{ hours}$$

So, you should aim for at least 9.28 hours of sleep.

Question 4:

Pep Zone sells auto parts and supplies including a popular multi-grade motor oil. The current replenishment policy: Place an order when the stock of this oil drops to 13 gal. During the 1-day lead-time, the demand follows Normal(10,3).

1. What is the probability of stock-out?
2. What should the reorder point be so that the probability of stock-out is 5%?

Answer:

Given the problem:

- Mean demand (μ) = 10 gallons
- Standard deviation (σ) = 3 gallons

1. Probability of stock-out:

The Z-score for a stock of 13 gallons is:

$$Z = \frac{13 - 10}{3} = 1$$

The probability of stock-out is:

$$P(X > 13) = 1 - P(Z \leq 1) = 1 - 0.8413 \approx 0.1587$$

So, the probability of stock-out is approximately 15.87%.

2. Reorder point for 5% probability of stock-out:

We need to find the Z-score corresponding to a 95% probability:

$$Z_{95\%} = 1.645$$

The reorder point is:

$$X = \mu + Z \cdot \sigma = 10 + 1.645 \cdot 3 \approx 14.935 \text{ gallons}$$

So, the reorder point should be approximately 14.935 gallons.

Question 5:

$n = 100, p = 0.4$. Find $P(30 \leq x \leq 50)$:

1. Using the binomial distribution, the result is
2. Using the normal approximation, $P(29.5 \leq x \leq 50.5) = ?$

Answer:

1. Using the binomial distribution:

Using the binomial distribution formula:

$$\begin{aligned} P(30 \leq x \leq 50) &= P(x \leq 50) - P(x \leq 29) \\ &= \text{Binom.dist}(50, 100, 0.4, 1) - \text{Binom.dist}(29, 100, 0.4, 1) = 0.98324 - 0.01478 = 0.968 \end{aligned}$$

2. Using the normal approximation:

The binomial distribution can be approximated by a normal distribution with mean μ and standard deviation σ , where:

$$\begin{aligned} \mu &= np = 100 \times 0.4 = 40 \\ \sigma &= \sqrt{np(1-p)} = \sqrt{100 \times 0.4 \times 0.6} = \sqrt{24} \approx 4.899 \end{aligned}$$

Using the continuity correction factor, we approximate:

$$P(30 \leq x \leq 50) \approx P(29.5 \leq x \leq 50.5)$$

We convert this to the standard normal distribution Z : $Z = \frac{X-\mu}{\sigma}$

For $x = 29.5$:

$$Z_{29.5} = \frac{29.5 - 40}{4.899} \approx -2.142$$

For $x = 50.5$:

$$Z_{50.5} = \frac{50.5 - 40}{4.899} \approx 2.141$$

Find the probabilities using the standard normal distribution table:

$$P(29.5 \leq x \leq 50.5) = P(-2.142 \leq Z \leq 2.141) = 0.9836 - 0.0164 = 0.967.$$